

ShopNow Enterprise Architecture — End-to-End Pipeline

A unified DevOps pipeline integrating GitHub, Jenkins, Terraform, Helm, and AWS EKS for secure, automated, and scalable application delivery.

◆ Extended Step-by-Step Flow

❏ Code & Build

- Developer commits → Jenkins CI/CD pipeline triggers.
- Jenkins builds immutable Docker images for each application (Frontend, Admin, Backend, MongoDB).
- Images are tagged and pushed to DockerHub (for portability/testing) and Amazon ECR (for AWS deployment).

❏ Infra Provisioning

- Terraform + Helm → VPC, EKS cluster, IAM roles, ELB setup.
- Helm charts reference ECR images for production, but can be swapped to DockerHub images for dev/test environments.

❏ Secret Management

- AWS Secrets Manager ↔ External Secrets Operator (IRSA).
- Kubernetes secrets for MongoDB credentials injected into pods securely.

❏ App Deployment

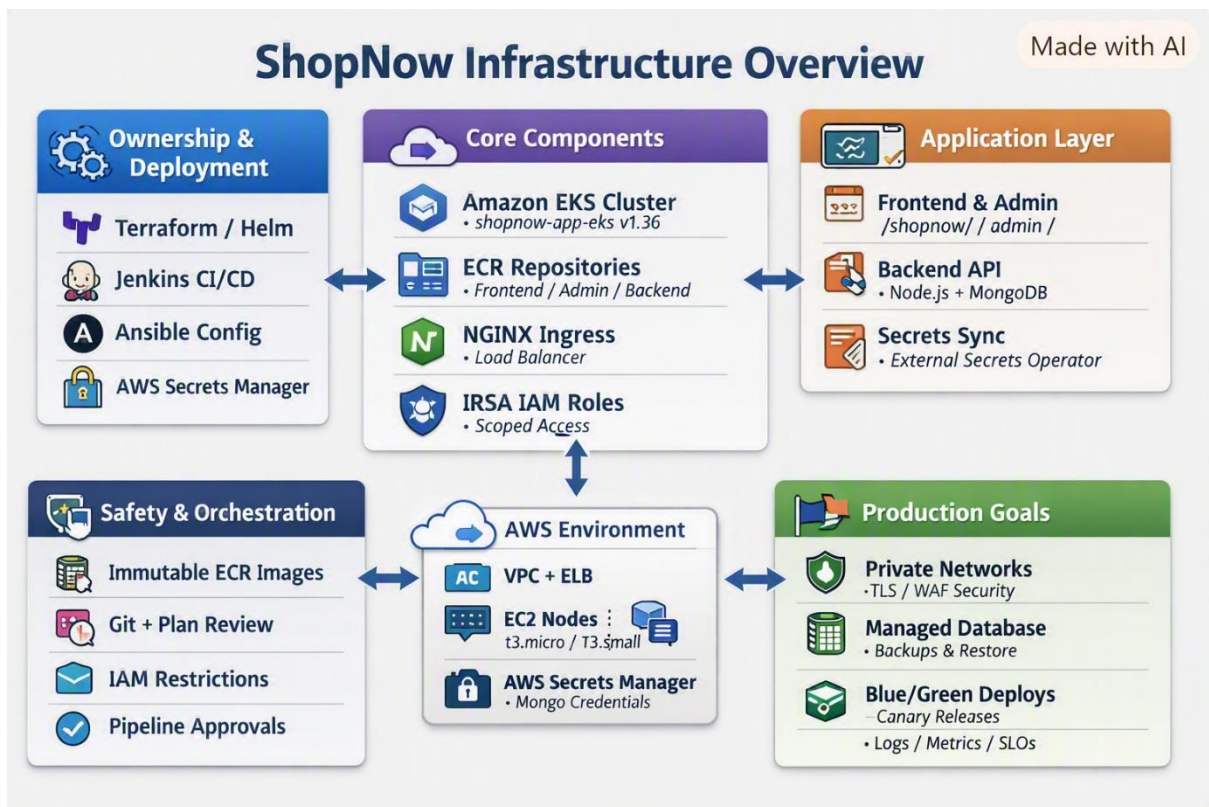
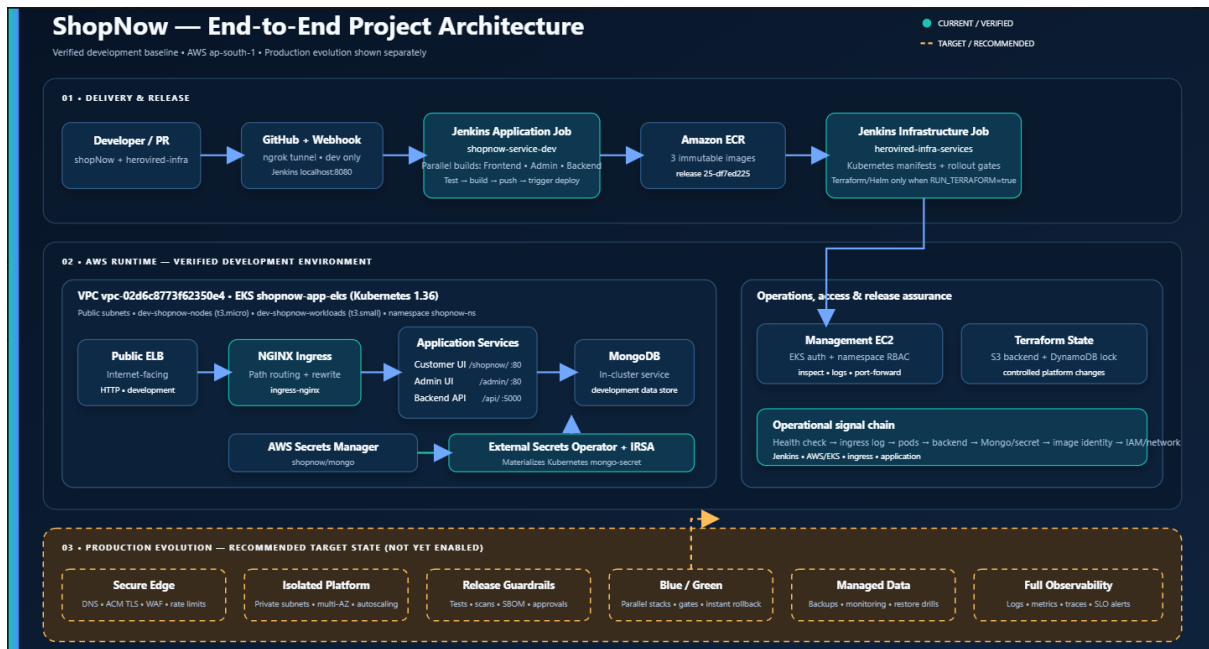
- Deploy Frontend, Admin, Backend pods via Helm.
- MongoDB StatefulSet/Service provisioned.
- Each pod pulls its image from ECR (prod) or DockerHub (dev/test).

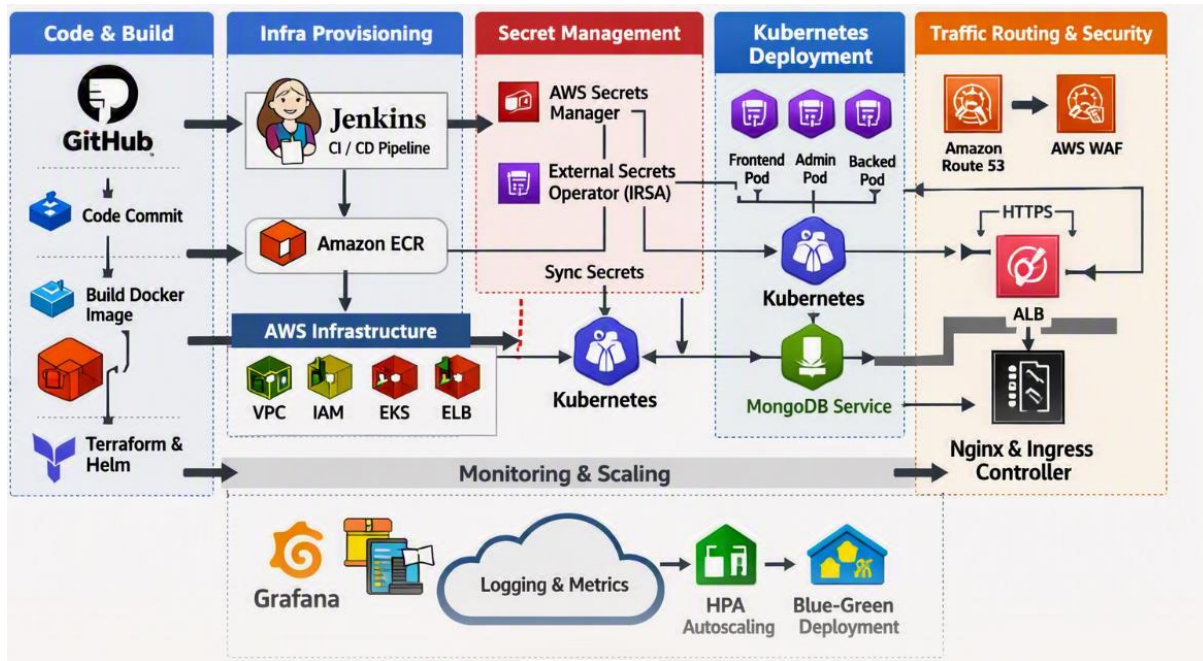
❏ Traffic Routing

- Route 53 + WAF → HTTPS ALB → NGINX Ingress Controller.
- Secure routing and filtering for external traffic.

❏ Monitoring & Scaling

- Centralized logs, metrics, alerts (CloudWatch, Prometheus, Grafana).
- HPA autoscaling based on CPU/memory.
- Blue/Green deployments supported via Helm upgrade strategies.





ShopNow AWS services currently used:

◆ Environment Overview

Category	Resource	Value
Account	AWS Account	559272000457
Region	Deployment Region	ap-south-1 (Mumbai)
Cluster	Amazon EKS	shopnow-app-eks (Kubernetes v1.36)
Namespace	Application Namespace	shopnow-ns
Ingress	Public Entry Point	Classic ELB → NGINX Ingress

◆ Compute & Networking

Component	Config	Capacity	Purpose	Status
Node Group – System	dev-shopnow-nodes (t3.micro)	2	General workloads	Current
Node Group – App	dev-shopnow-workloads (t3.small)	2 desired, max 3	Application workloads	Current
Management EC2	i-05988c1864c27c07f	1 instance, IP 13.201.136.220	Cluster admin	Current
VPC	vpc-02d6c8773f62350e4	1	Network boundary	Current
Subnets	subnet-088ea34a6270f1000 , subnet-04b729189cb399e38	2	Multi-AZ connectivity	Current

◆ Containers & Release

Component	Resource	Purpose	Status
ECR Repositories	shopnow-dev/frontend , shopnow-dev/admin , shopnow-dev/backend	Store Docker images	Current
Release Identity	Tag 25-df7ed225	Immutable app release	Current
CI/CD Jobs	Jenkins shopnow-service-dev , herovired-infra-services	Build & deploy	Current

◆ Secrets & Identity

Component	Resource	Purpose	Status
Secrets Manager	shopnow/mongo	MongoDB credentials	Current
External Secrets Operator	Helm chart external-secrets-2.9.0	Sync AWS → K8s	Current
IRSA Role	Scoped IAM role	Grants ESO access	Current

◆ State & Configuration

Component	Resource	Purpose	Status
Terraform Backend	S3 <code>harish-pc-s3-bucket</code>	State storage	Current
Terraform Locking	DynamoDB <code>shopnow-terraform-locks</code>	Prevents concurrent ops	Current
Helm Releases	NGINX, External Secrets	Cluster add-ons	Current
Ansible	Management playbooks	EC2 config	Conditional

Kubernetes Application Inventory

Namespace	Service	Port/Route	Purpose	Dependency
<code>shopnow-ns</code>	Frontend	<code>80</code> → <code>/shopnow/</code>	Customer portal	Backend API
<code>shopnow-ns</code>	Admin	<code>80</code> → <code>/shopnow/admin/</code>	Admin portal	Backend API
<code>shopnow-ns</code>	Backend	<code>5000</code> → <code>/shopnow/api/</code>	Business logic	MongoDB
<code>shopnow-ns</code>	MongoDB	Internal	Dev DB	<code>mongo-secret</code>
<code>shopnow-ns</code>	External Secrets Operator	Internal	Sync secrets	AWS Secrets Manager
<code>ingress-nginx</code>	Ingress Controller	Public ELB	Routes requests	All services

Compute Capacity Summary

Resource	Type	Min	Desired	Max	Current
Node Group – System	<code>t3.micro</code>	2	2	2	2
Node Group – App	<code>t3.small</code>	2	2	3	2
Management EC2	—	—	1	—	1
Total EC2-backed capacity	Mixed	—	—	—	5 instances

Public Endpoints

Application	Path	Protocol	Readiness
Customer Portal	/shopnow/	HTTP	Dev only
Admin Portal	/shopnow/admin/	HTTP	Dev only
Backend API	/shopnow/api/	HTTP	Dev only
API Health	/shopnow/api/health	HTTP	Dev only

Current vs. Production Target

Area	Current	Recommended Production
Environment isolation	Single dev account	Separate dev/stage/prod accounts
Networking	Public subnets	Private multi-AZ subnets
Public access	HTTP ELB	DNS + TLS + HTTPS + WAF
Database	In-cluster MongoDB	Managed DB (DocumentDB/Atlas)
Deployment	Rolling rollout	Blue/green or canary
CI authentication	Jenkins app creds	Short-lived CI IAM role
Image security	Immutable tags	Scanning + SBOM + provenance
Scaling	Fixed nodes	Autoscaler + HPA
Monitoring	CLI & pod logs	Centralized logs/metrics/traces
Resilience	Dev baseline	PDBs, probes, restore drills

AWS Side screenshots:

EKS + Node Group

The screenshot shows the AWS Management Console for the 'shopnow-app-eks' cluster. The 'Nodes' tab is active, displaying a table of nodes and their associated node groups. The table includes columns for Node name, Instance type, Compute, Managed by, Created, Status, CPU usage, Memory usage, and Ephemeral storage usage. The node groups are listed below the table, showing their names, desired sizes, AMI release versions, launch templates, and statuses.

Node name	Instance type	Compute	Managed by	Created	Status	CPU usage	Memory usage	Ephemeral storage usage
ip-10-20-1-162.ap-south-1.compute.internal	t3.small	Node group	dev-shopnow-workloads	21 hours ago	Ready	23%	40%	15%
ip-10-20-1-25.ap-south-1.compute.internal	t3.micro	Node group	dev-shopnow-nodes	August 18, 2026, 00:39 (UTC+05:30)	Ready	22%	59%	15%
ip-10-20-2-142.ap-south-1.compute.internal	t3.micro	Node group	dev-shopnow-nodes	a day ago	Ready	17%	58%	15%
ip-10-20-2-146.ap-south-1.compute.internal	t3.small	Node group	dev-shopnow-workloads	21 hours ago	Ready	17%	30%	15%

Group name	Desired size	AMI release version	Launch template	Status
dev-shopnow-nodes	2	1.36.2-20260810	-	Active
dev-shopnow-workloads	2	1.36.2-20260810	-	Active

ECR

The screenshot shows the AWS Management Console for the Amazon Elastic Container Registry (ECR). The 'Private repositories' section is active, displaying a table of repositories. The table includes columns for Repository name, URI, Created at, and Tag immutability. The repositories listed are 'shopnow-dev/admin', 'shopnow-dev/backend', and 'shopnow-dev/frontend'.

Repository name	URI	Created at	Tag immutability
shopnow-dev/admin	559272000457.dkr.ecr.ap-south-1.amazonaws.com/shopnow-dev/admin	16 August 2026, 13:40:17 (UTC+05:5)	Immutable
shopnow-dev/backend	559272000457.dkr.ecr.ap-south-1.amazonaws.com/shopnow-dev/backend	16 August 2026, 13:42:53 (UTC+05:5)	Immutable
shopnow-dev/frontend	559272000457.dkr.ecr.ap-south-1.amazonaws.com/shopnow-dev/frontend	16 August 2026, 13:43:17 (UTC+05:5)	Immutable

EC2

The screenshot shows the AWS Management Console for the Amazon Elastic Compute Cloud (EC2). The 'Instances' section is active, displaying a table of instances. The table includes columns for Name, Instance ID, Instance state, Instance type, Status check, Availability Zone, Public IPv4 DNS, Public IPv4, and Elastic IP. The instances listed are 'dev-shopnow-management-instance' and 'dev-shopnow-management-instance'.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4	Elastic IP
	i-0d0134ad00ac09e9b	Running	t3.micro	3/3 checks passed	ap-south-1b	ec2-13-200-215-49.ap...	13.200.215.49	-
	i-063fe76f767b12e10	Running	t3.small	3/3 checks passed	ap-south-1b	ec2-43-204-150-69.ap...	43.204.150.69	-
dev-shopnow-management-instance	i-0dabec0e0511f1ae	Running	t3.micro	3/3 checks passed	ap-south-1a	ec2-13-207-55-28.ap...	13.207.55.28	-
	i-0951a99a069300c71	Running	t3.micro	3/3 checks passed	ap-south-1a	ec2-3-6-92-151.ap-sou...	3.6.92.151	-
dev-shopnow-management-instance	i-05988c1864c27c07f	Running	t3.micro	3/3 checks passed	ap-south-1a	ec2-13-201-136-220.ap...	13.201.136.220	-
	i-03dbfaf06b58892a	Running	t3.small	3/3 checks passed	ap-south-1a	ec2-13-233-114-186.ap...	13.233.114.186	-

Classic Load Balancer:

DNS name:

a2d7eee8d8179427fa36d881be68d64a-277526266.ap-south-1.elb.amazonaws.com

EC2

Dashboard

AWS Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Trust Stores

Auto Scaling

Auto Scaling Groups

a2d7eee8d8179427fa36d881be68d64a

▼ Details

Load balancer type

Classic

Status

4 of 4 instances in service

VPC

vpc-02d6c8773f62350e4

Date created

August 19, 2026, 15:17 (UTC+05:30)

Scheme

Internet-facing

Hosted zone

ZP97RAFLXTNZK

Availability Zones

subnet-088ea34a6270f1000

ap-south-1a (aps1-az1)

subnet-04b729189cb399e38

ap-south-1b (aps1-az3)

DNS name

a2d7eee8d8179427fa36d881be68d64a-277526266.ap-south-1.elb.amazonaws.com

(A Record)

This Classic Load Balancer can be migrated to a next-generation load balancer. Migration wizard uses your load balancer's current configurations to create a new load balancer. [Learn more](#)

Launch migration wizard

Distribution of targets by Availability Zone (AZ)

For each enabled Availability Zone, you can view the number of registered instances and their current health states. Selecting any values here will apply the corresponding filter to the Target instances table.

Listeners

Network mapping

Security

Health checks

Target instances

Monitoring

Attributes

Tags

Listeners

A Classic Load Balancer listener uses the protocols and ports you configure, to both check for connection requests and forward received traffic to instances. The listener uses its protocol and port to check for connection requests. When traffic is received by the listener, it's forwarded to registered EC2 instances using the instance protocol and port.

Q Filter listeners

Protocol:Port	Instance Pr...	Security policy	Default SSL/TLS certificate	Cookie stickiness	Backend server description
TCP:443	TCP:32701	Not applicable	Not applicable	Not applicable	Not applicable
TCP:80	TCP:32032	Not applicable	Not applicable	Not applicable	Not applicable

Listeners

Network mapping

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Health checks

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Attributes

Tags

Network mapping

Target instances in the listed Availability Zones and subnets are available to receive traffic from the load balancer.

VPC

vpc-02d6c8773f62350e4

IPv4 VPC CIDR: 10.20.0.0/16

IPv6 VPC CIDR: -

Availability Zones and subnets

Including two or more Availability Zones, and corresponding subnets, increases the fault tolerance of your applications.

Zone	Subnet	IPv4 address	Private IPv4 address
ap-south-1a (aps1-az1)	subnet-088ea34a6270f1000	Assigned by AWS	Assigned from CIDR 10.20.1.0/24
ap-south-1b (aps1-az3)	subnet-04b729189cb399e38	Assigned by AWS	Assigned from CIDR 10.20.2.0/24

Listeners

Network mapping

Security

Health checks

Target instances

Monitoring

Attributes

Tags

Security groups (1)

A security group is a set of firewall rules that control the traffic to your load balancer.

Security Group ID	Name	Description
sg-0bd2c2b970bb8fe44	k8s-elb-a2d7eee8d8179427fa36d881be68d64a	Security group for Kubernetes ELB a2d7eee8d8179427fa36d881be68d64a (ingress-nginx/ingress-nginx-controller)

Listeners

Network mapping

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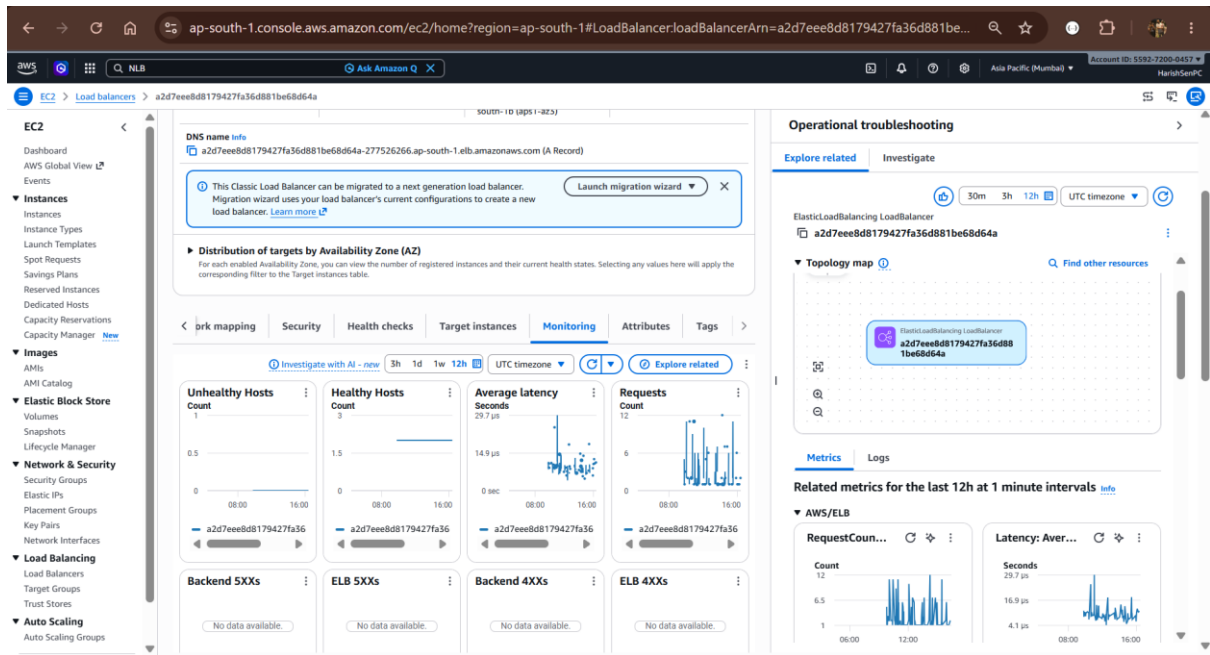
Target instances (4)

Instances currently registered to your load balancer are displayed. To deregister instances, select them, then choose Deregister. To register and deregister instances simultaneously, choose Manage instances.

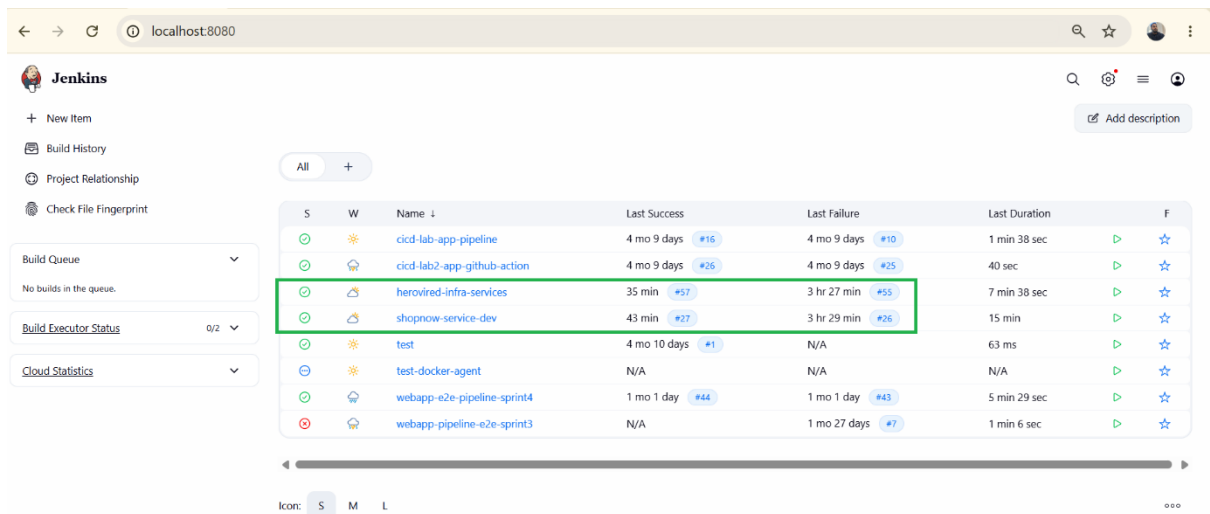
Q Filter target instances

	Instance ID	Name	Health status	Health status descri...	Security groups	Zone	Public IPv4 address
☐	i-0d0134ad00ac09e9b		In-service	Not applicable	eks-cluster-sg-shopnow-app-eks-8504...	ap-south-1b	13.200.215.49
☐	i-063fe76f767b12e10		In-service	Not applicable	eks-cluster-sg-shopnow-app-eks-8504...	ap-south-1b	43.204.150.69
☐	i-0951a99a069300c71		In-service	Not applicable	eks-cluster-sg-shopnow-app-eks-8504...	ap-south-1a	3.6.92.151
☐	i-03dbfaf06b588992a		In-service	Not applicable	eks-cluster-sg-shopnow-app-eks-8504...	ap-south-1a	13.233.114.186

Grafana:

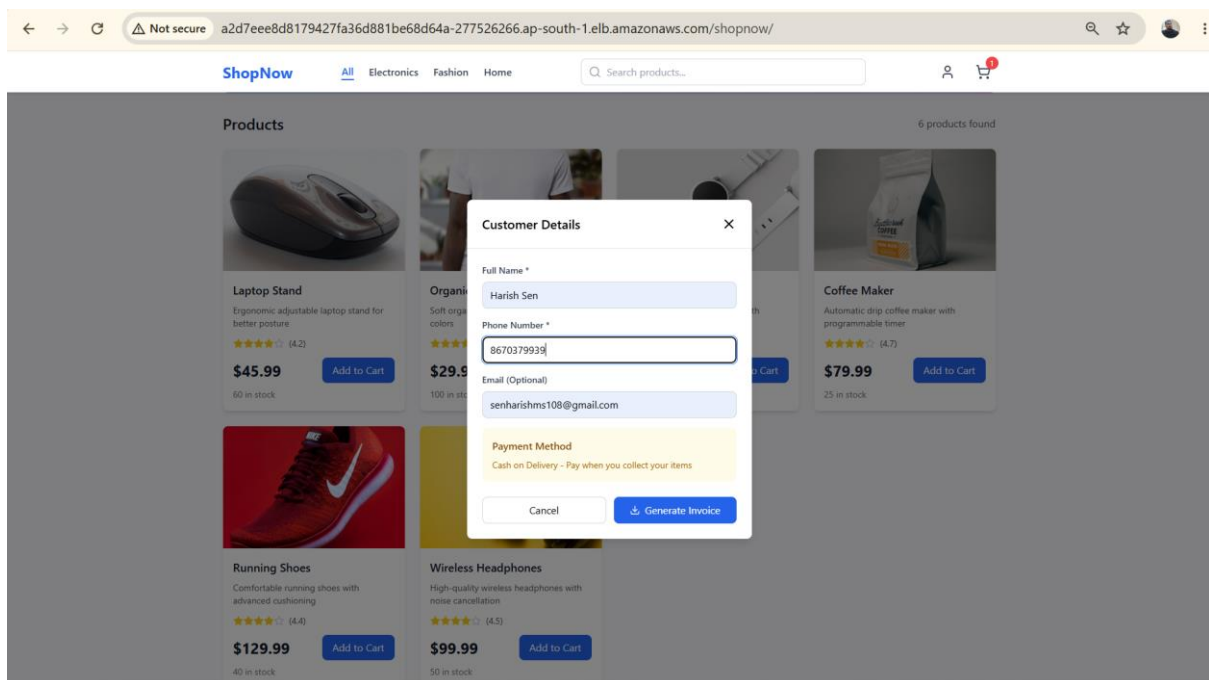
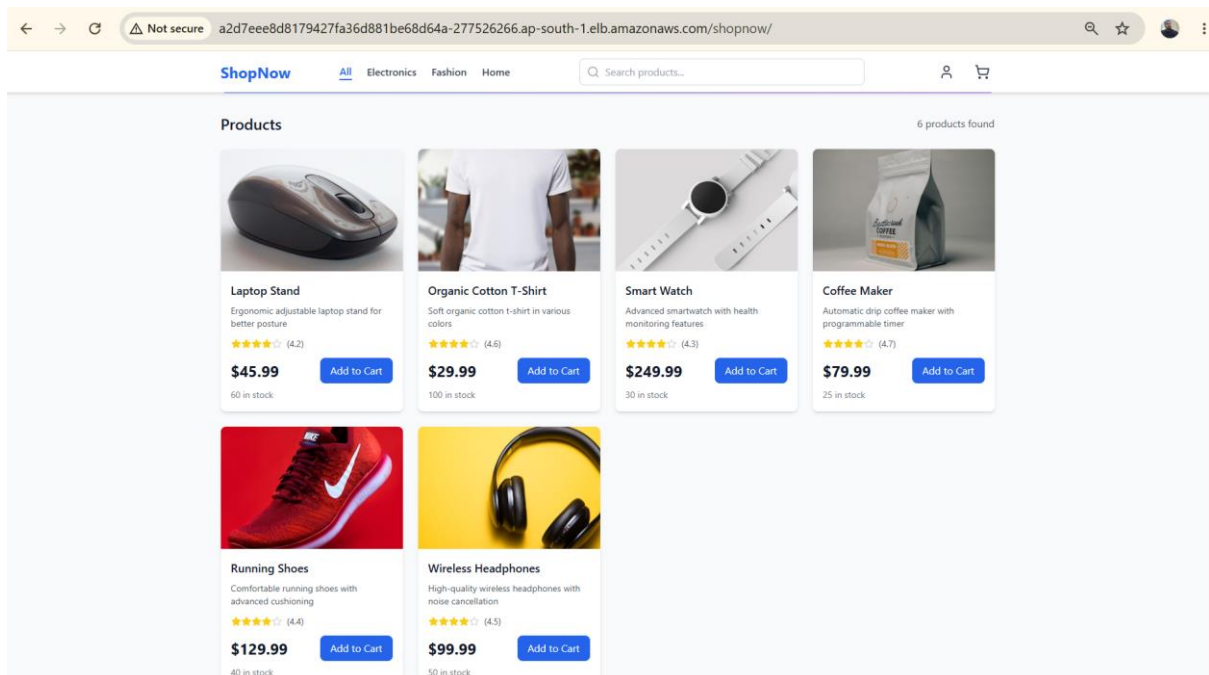


Jenkins Pipeline:



Access Application via Load balancer URLs:

<http://a2d7eee8d8179427fa36d881be68d64a-277526266.ap-south-1.elb.amazonaws.com/shopnow/>



<http://a2d7eee8d8179427fa36d881be68d64a-277526266.ap-south-1.elb.amazonaws.com/shopnow/admin/>

← → ↻ ⚠ Not secure a2d7eee8d8179427fa36d881be68d64a-277526266.ap-south-1.elb.amazonaws.com/shopnow/admin/ 🔍 ☆ 👤 ⋮

ShopNow Admin

Order Management Dashboard

Today's Orders **1**

Total Orders
1

Pending
1

Ready
0

Total Revenue
\$0.00

Orders

All Status ▾

ORDER DETAILS	CUSTOMER	TOTAL	STATUS	DATE	ACTIONS
SN1787137254047592 Token: 274753	Harish Sen 6670379933	\$45.99	Pending	8/19/2026	Mark Ready

<http://a2d7eee8d8179427fa36d881be68d64a-277526266.ap-south-1.elb.amazonaws.com/shopnow/api/health>

← → ↻ ⚠ Not secure a2d7eee8d8179427fa36d881be68d64a-277526266.ap-south-1.elb.amazonaws.com/shopnow/api/health 🔍 ☆ 👤 ⋮

pretty-print ☐

```
{
  "status": "OK",
  "message": "ShopNow API is running"
}
```

DB Level checks:

```
harish@Harish:/tmp$ kubectl get pods,svc -n shopnow-ns -l app=mongo
NAME                                READY   STATUS    RESTARTS   AGE
pod/mongo-bc4cdbf9f-jbsz4          1/1     Running   0           13h

harish@Harish:/tmp$ MONGODB_URI=$(kubectl get secret mongo-secret \
  -n shopnow-ns \
  -o jsonpath='{.data.MONGODB_URI}' | base64 --decode)
harish@Harish:/tmp$ echo $MONGODB_URI
mongodb://shopuser:ShopNowPass123@mongo:27017/shopnow?authSource=admin
harish@Harish:/tmp$ mongosh "$MONGODB_URI"

harish@Harish:/tmp$ kubectl exec -it -n shopnow-ns deploy/mongo -- mongosh "$MONGODB_URI"
Current Mongosh Log ID: 6a85e85934068b659dc2cf3c
Connecting to:      mongodb://<credentials>@mongo:27017/shopnow?authSource=admin&directConnection=true&appName=mongosh+2.10.0
Using MongoDB:      7.0.40
Using Mongosh:      2.10.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

shopnow> use shopnow
already on db shopnow
shopnow> show collections
invoices
products
users
```

```
shopnow> db.products.find().pretty()
[
  {
    _id: ObjectId('6a8556d4b03773879eb2ade0'),
    name: 'Wireless Headphones',
    price: 99.99,
    category: 'electronics',
    image: 'https://images.unsplash.com/photo-1505740420928-5e560c06d30e?w=300&h=300&fit=crop&crop=center',
    description: 'High-quality wireless headphones with noise cancellation',
    rating: 4.5,
    stock: 50,
    createdAt: ISODate('2026-08-19T07:10:12.403Z'),
    __v: 0
  },
  {
    _id: ObjectId('6a8556d4b03773879eb2ade1'),
    name: 'Smart Watch',
    price: 249.99,
    category: 'electronics',
    image: 'https://images.unsplash.com/photo-1523275335684-37898b6baf30?w=300&h=300&fit=crop&crop=center',
    description: 'Advanced smartwatch with health monitoring features',
    rating: 4.3,
    stock: 30,
    createdAt: ISODate('2026-08-19T07:10:12.404Z'),
    __v: 0
  },
]
```

```
shopnow> db.invoices.find().pretty()
[
  {
    _id: ObjectId('6a858ce6a048db4ad4c24962'),
    invoiceNumber: 'SN1787137254047592',
    token: '274753',
    customerDetails: {
      name: 'Harish Sen',
      phone: '8670379939',
      email: 'senharishms108@gmail.com'
    },
    items: [
      {
        _id: '6a8556d4b03773879eb2ade4',
        name: 'Laptop Stand',
        price: 45.99,
        quantity: 1,
        image: 'https://images.unsplash.com/photo-1527864550417-7fd91fc51a46?w=300&h=300&fit=crop&crop=center'
      }
    ],
    total: 45.99,
    status: 'pending',
    paymentMethod: 'Cash on Delivery',
    paymentStatus: 'pending',
    createdAt: ISODate('2026-08-19T11:00:54.061Z'),
    __v: 0
  }
]
```

```
shopnow> db.users.find().pretty()
```

```
shopnow>
```